ETE4351L Data Communications and Networking

Lab 4: Understanding PWM and demodulation

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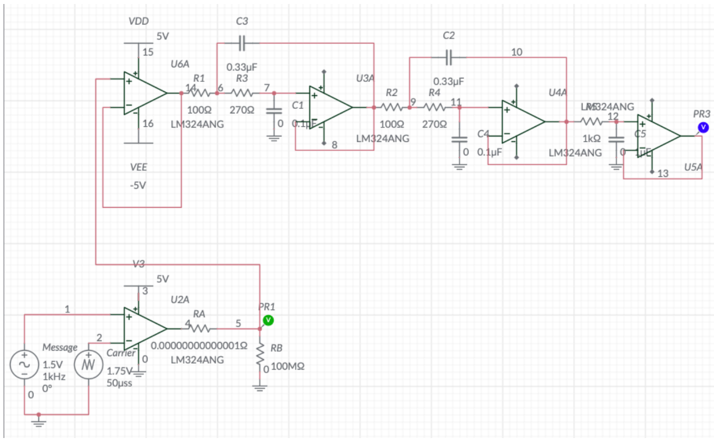
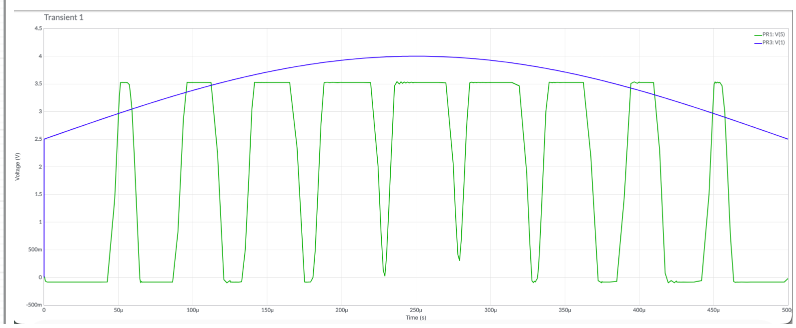




Objectives:

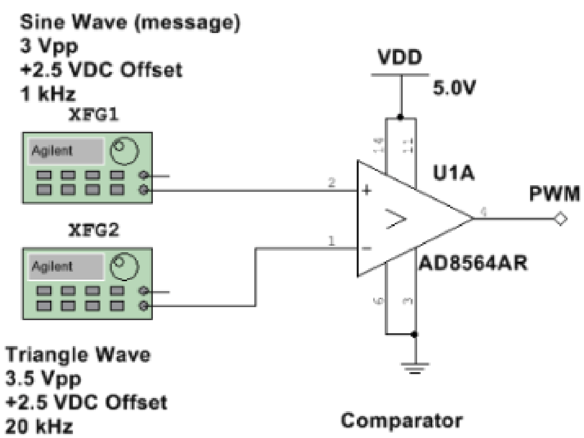
- Examine common signals for frequency content
- Use computer software such as MATLAB to aid in calculations
- Reinforce use of lab equipment such as a spectrum analyzer

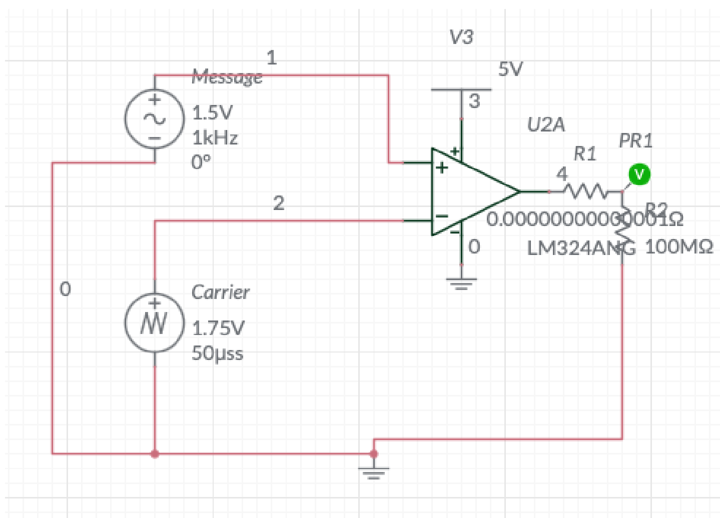
Introduction

PWM (pulse width modulation) is a way to send analog information using *only digital voltage levels* (high and low voltages). PWM can be used to send information and can be used very similarly to how DAC (digital to analog converter) is used. Instead of changing the amplitude of a signal PWM changes the width of the pulses (also known as the duty cycle) to carry information. The duty cycle determines how long the signal stays on as compared to how long it stays off during each cycle, which directly affects the average voltage of a signal. In this lab, a PWM signal is generated using a comparator by comparing a low frequency sine wave (the message signal) with a high frequency triangle wave. The output of the comparator produces a pulse signal whose duty cycle varies based on the amplitude of the sine wave. This demonstrates how analog information can be converted into a digital form.

This lab also covers how to recover the original analog signal from the PWM waveform, using a low pass filter. By removing the high frequency components of the PWM signal the filter allows the average value of the waveform to pass through. This closely resembles the original input signal, and this process is like demodulation. Overall this lab aims to demonstrate how PWM can be used to encode and decode analog signals, and highlights its importance in real world applications such as signal transmission and digital to analog conversion.

Procedure:



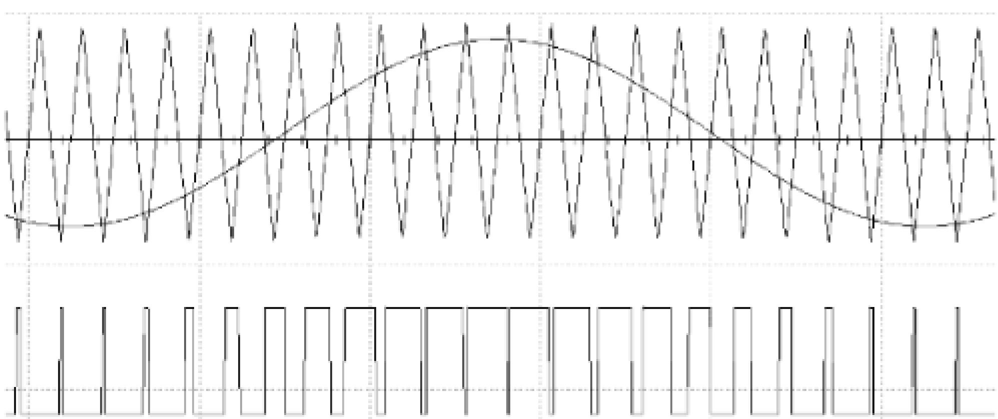


IMG 1. *PWM Generator with Comparator*

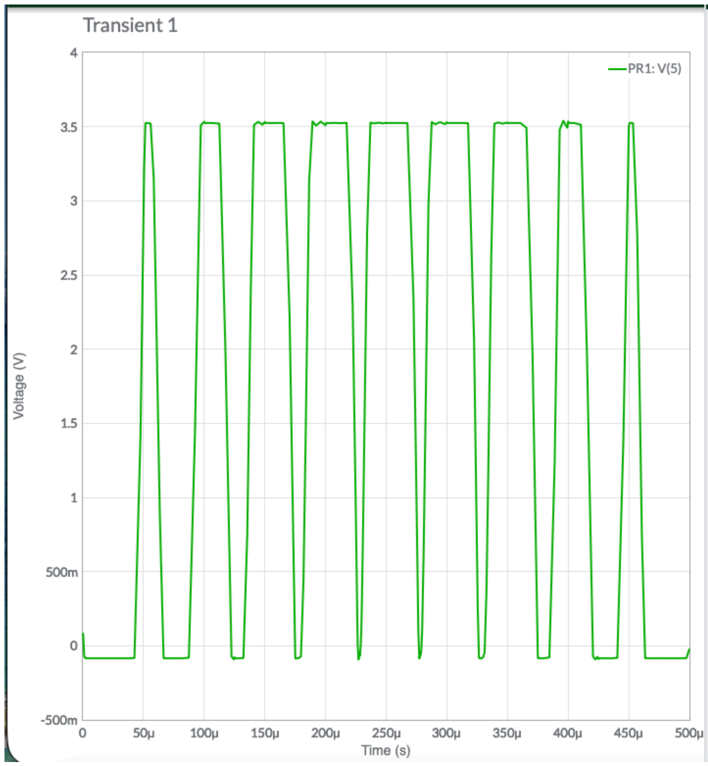
IMG 2. *PWM Generator with Comparator, built on multisim. The circuit can be found here:* https://www.multisim.com/content/ThCqqnix3MoeoZALWvwBRw/lab-4-circuit-from-image-1/open/

The message signal is the information that is being sent. The triangle wave generator is used to provide a changing reference value that the sine wave message signal can then be compared to. Whenever the voltage of the message sine wave is larger than the triangle wave, the output of the comparator will be high. This is because this is the case where the + pin (non inverting) is above the inverting (-) pin of the comparator. However, if the message amplitude was increased beyond the triangle wave's amplitude, pulses would likely be missed because the sine wave would not cross the triangle for a few cycles, therefore no pulses would be generated at that time.

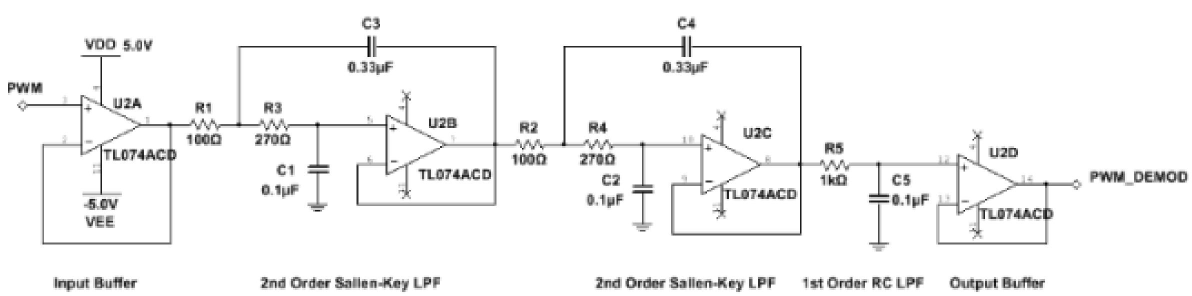
As far as specifics about the multisim circuit in figure two, since $f = \frac{1}{t}$, $so\ 20,000Hz = \frac{1}{0.00005}$ and the period is 0.00005 seconds. There are also a few small tweaks made to the circuit to make it able to simulate, such as adding a very high resistance value to ground (so minimal current is going at this branch) and a very small resistor at the output of the comparator. Note that the voltages are not vpp hence why the triangle wave voltage is entered as 1.75V and the sine wave signal is entered as 1.5V.



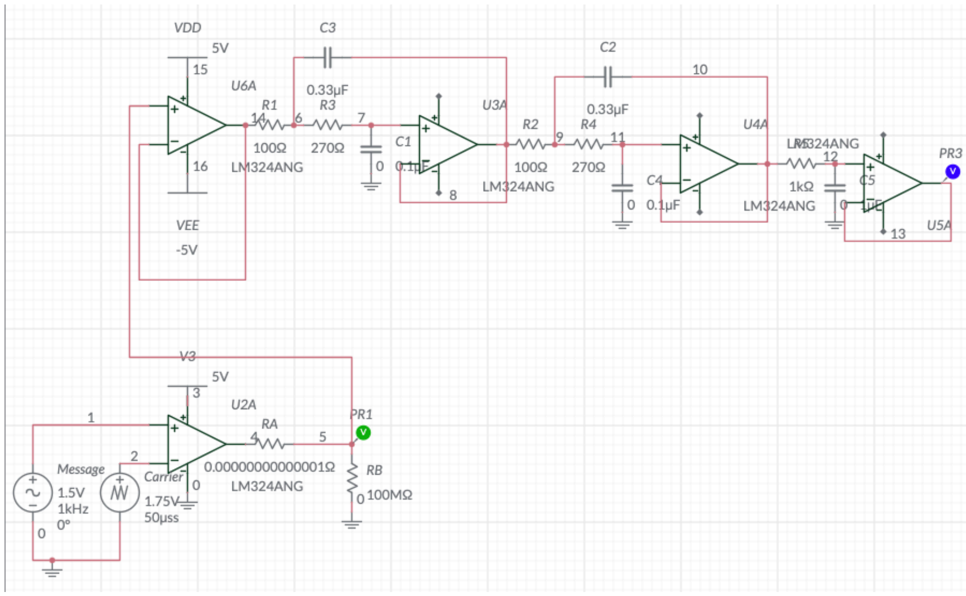
IMG 3. *Example PWM output*



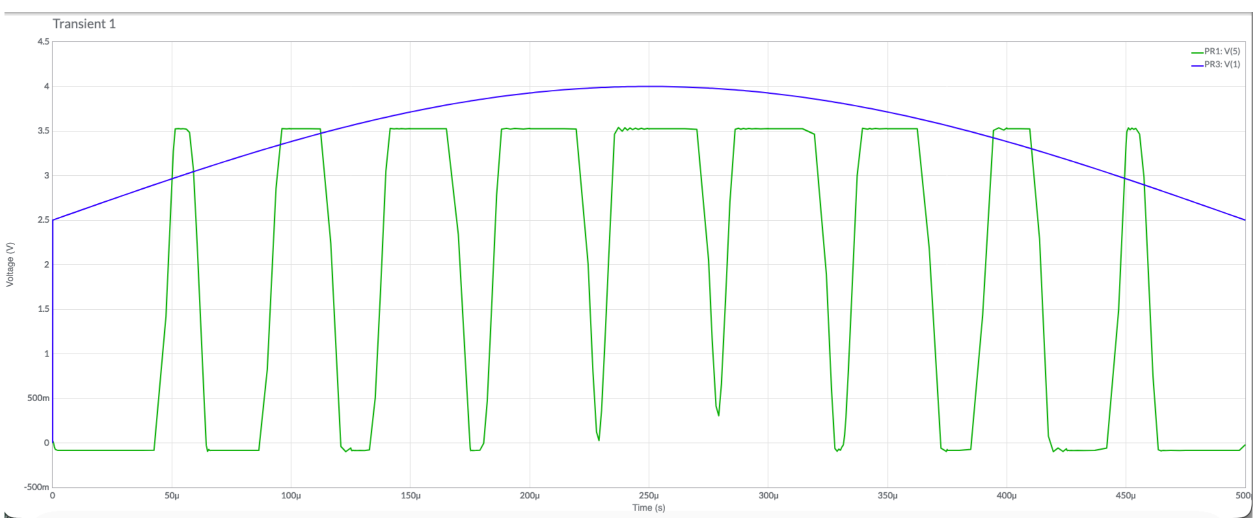
IMG 4. *Our observed PWM output*



IMG 5. *PWM demodulator circuit*



IMG 6. *PWM demodulator circuit on Multisim, found here:* [*https://www.multisim.com/content/MezMigsXMJB8mJbwDFMqaU/lab-4-circuit-2/open/*](https://www.multisim.com/content/MezMigsXMJB8mJbwDFMqaU/lab-4-circuit-2/open/)



IMG 7. *PWM demodulator circuit results on Multisim*

1.In your report explain the expected behavior of the PWM output based on the operation of the comparator and input signals.

It can be seen in image 4. that the PWM output switches between low and high voltage levels. The pulse width changes over time based on the amplitude of the sine wave message signal, as we can see that the output switches between low and high and the pulse width changes. The high level is around 3.5 volts. When the sine wave was greater than the triangle wave the comparator output went high, and when the sine wave was less than the triangle wave the output went low. This confirmed that the comparator was generating PWM.

2.In the demodulated circuit, which frequency ranges do we desire to pass through the filter? What frequencies do we need to block?

The demodulator circuit consists of input and output buffers as well as some filter stages in between. This filter was designed knowing that we need to understand the frequency components of our signals to decide how to best design the filter (fourier series). We can see we need a low pass filter. Every time the duty cycle of the PWM signal changes the average DC value changes with it. The DC component has a frequency of 0 but the frequency of the sinusoidal message carrier is 1khz (img 6). The pulses are generated at a frequency of 20kHz, so we need our filter to pass the DC component and message signal at 1kHz but also blocking the high frequency pulses at 20kHz. This is why we need a low pass filter to accomplish this.

3.What is the order of the LPF used in the demodulator circuit? What would be affected by reducing the demodulating filter? Given the op amp circuit used to generate the PWM signal, why do the input sine and triangle signals need DC offset?

The low pass filter has two stages of Salen Key filters and one stage of RC. So therefore the LPF in the demodulator circuit is 5th order. If we reduce the filter order the filter becomes less effective at removing high frequency components because the attenuation past the critical frequency is less severe of a drop off. This means more of the PWM switching signal at 20kHz will pass through instead of being blocked. As a result the output will have more noise and ripple and will not look as smooth as the original sine wave. Essentially, the more orders of the filter the more visually similar the output wave will look.

The sine and triangle signals need a DC offset so that they remain positive and within the operating range of the op amp. If the signals go negative, the op amp (used as a comparator) may not work properly because it cannot handle voltages outside its allowed range. The DC offset shifts both signals upward so that they can be compared properly and allow the PWM signal to be generated properly.

Conclusion

In this lab, we created a PWM signal and saw how the duty cycle changes based on the input sine. We then used a demodulator to get back the original sine signal, and used a low pass filter to help recover. This lab showed us how PWM can be used to convert between analog and digital signals. Overall, the lab helped us understand how PWM works and why filtering is important for getting a clean output after demodulation.